Lyceum helps lecturers keep their courses in step with what industry needs.

It is an assistant for university lecturers. A small team of AI agents keeps watch over each subject: they gather what employers are asking for from job sites, report what has changed, and, once a change is approved, help the lecturer update the course and build what is needed to teach it, from slides to test questions and quizzes that fit the Course Learning Outcomes (CLOs). After the test, the results come back in, so the next review starts from evidence instead of guesswork.

WHY IT MATTERS

MALAYSIA

32.2%

of graduates worked below their qualification in 2024, while graduate unemployment was only 3.2%

DOSM, Graduates Statistics 2024

ASIA PACIFIC

77%

of employers can't find the skills they need, up from 45% in 2014

ManpowerGroup, Talent Shortage 2025 (10,095 employers surveyed)

WORLD

39%

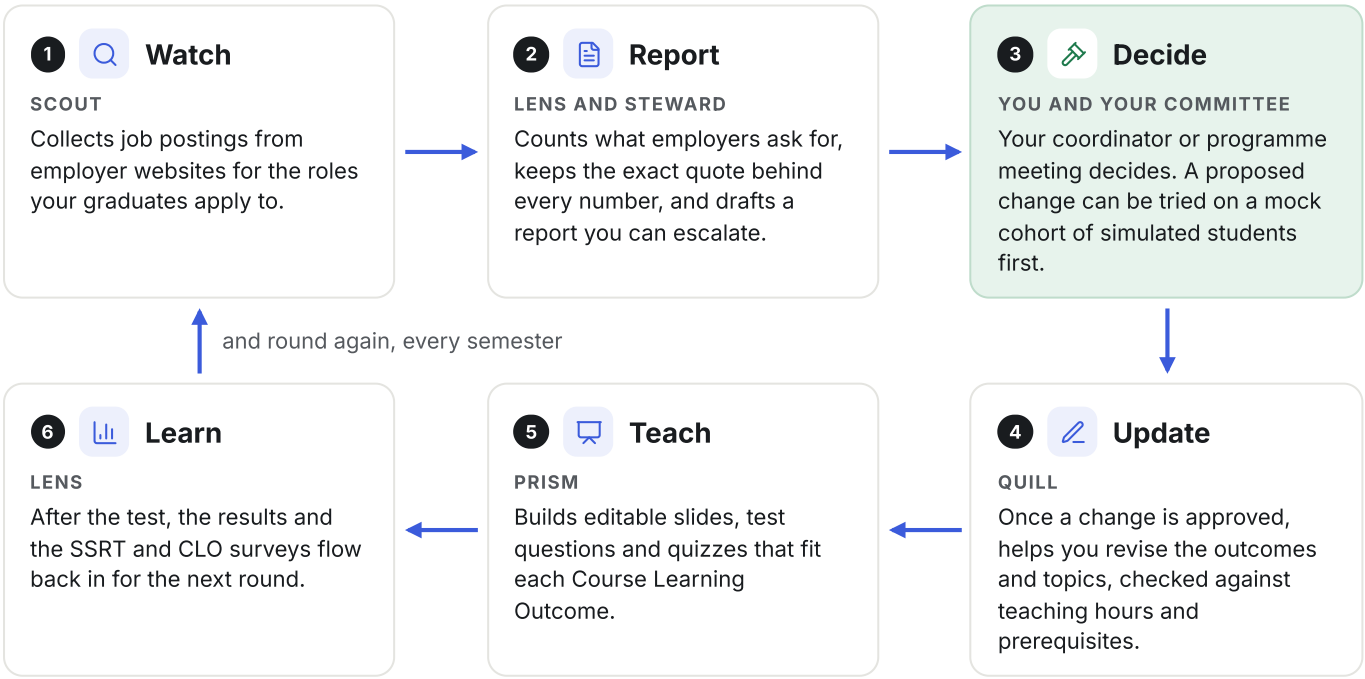
of workers' skills are expected to change or become outdated by 2030

World Economic Forum, Future of Jobs Report 2025

Graduates are finding jobs. The problem is the match between what they learned and what the job asks for. Courses are reviewed only every few years, and the evidence behind a review usually sits in browser tabs and people's memory.

HOW IT WORKS

One loop, six steps, and people decide



Seven agents share the work, each with a name and a job: **Folio** reads the course, **Scout** watches industry, **Atlas** maps skills to outcomes, **Lens** counts, **Quill** designs, **Prism** makes the materials and **Steward** prepares the review. Nothing changes a course until a person approves it.

Sources: DOSM, Graduates Statistics 2024; ManpowerGroup, Talent Shortage 2025, Asia Pacific; World Economic Forum, Future of Jobs Report 2025.

WHAT MAKES IT DIFFERENT

PLANNED

Fits the tools you already use
Lyceum isn't one more app to log into. It is built to sit inside Microsoft 365 and Teams, Google Workspace, the LMS and the university's own AI assistant, through MCP and extensions.

WORKING

Every number has a receipt
Each requirement is an exact quote from a real job posting, and every count shows what it is out of, so any number can be traced back to its source.

WORKING

The lecturer stays in charge
Agents gather, draft and check. People decide, and the university's own approval process stays in control.

WHERE IT STANDS TODAY

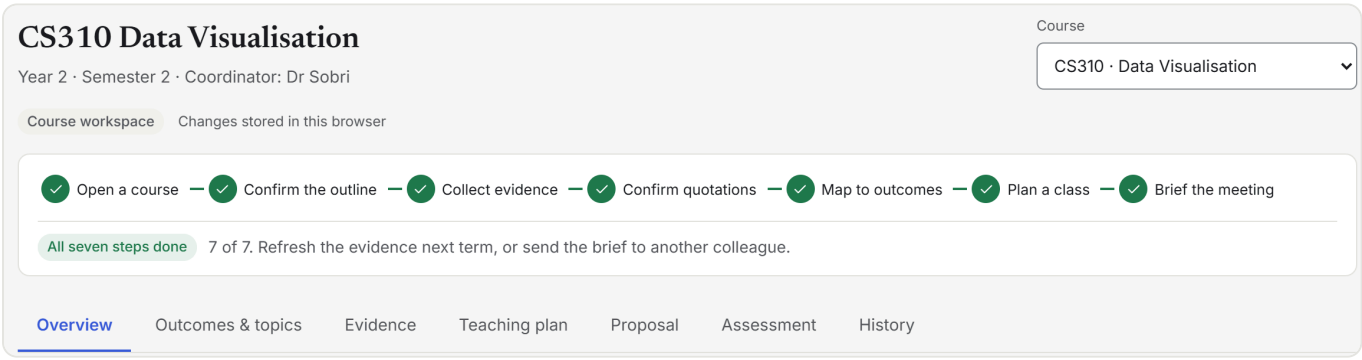
Working now, as a local prototype

- Pulls real job postings from employer job boards (Greenhouse and Lever) when the lecturer asks
- Checks every quote against its source and counts honestly
- Prepares the report and sends it to a colleague inside the app
- Updates the course outline, checked against hours and prerequisites
- Exports editable PowerPoint slide packs and drafts template quiz items against the CLOs
- 160 automated checks pass on the current code

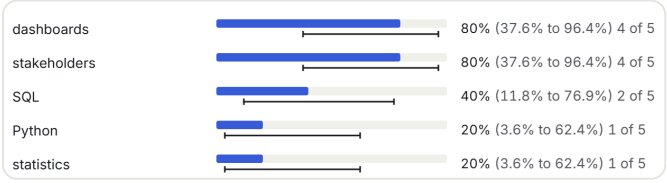
Still to build

- Watching on a schedule (today it runs when asked)
- The MCP, Microsoft 365, Google and LMS plug-ins
- AI-drafted test questions and quizzes
- A mock cohort benchmarked against real survey results
- Mascots and personas for the agents

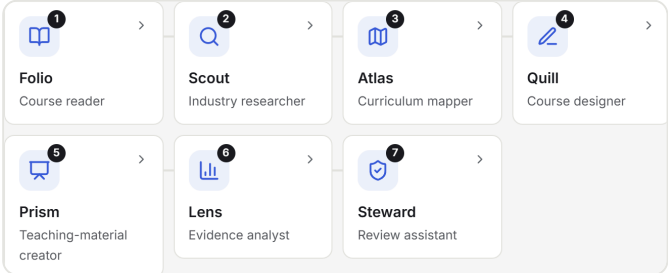
WHAT IT LOOKS LIKE



The course workspace. Its guided steps tick off only when the real work behind them is saved. They will be renamed to match the six-step loop on page 1, which is one of the review's points.



The report. How many of the reviewed postings ask for each skill. The thin line under each bar shows the uncertainty range.



The seven agents. Each card opens to show what the agent reads, produces and cannot do. Mascots and personalities come next.

Real captures of the working prototype, filled by its built-in demo with a saved sample of real job postings.

REVIEW

Comments from Cai Yiyu

Cai Yiyu is the organizer who approves the travel fellowships, and had already approved another team's. For ours to be approved as well, I presented Lyceum so Cai Yiyu could judge whether the project was worth sponsoring. These were the comments.

WHAT LANDED WELL

Solid, clear, on theme

The idea is solid and clear, and it fits the theme.

The platform answer

Asked why a lecturer would pick Lyceum over other apps, especially when they already juggle too many tools, I said we are not selling one more separate tool. Lyceum is a platform that fits into any university's existing ecosystem, through MCP and extensions for Microsoft 365, Google and the AI tools they already use. Cai Yiyu liked the concept. It isn't built yet, so it is now a committed work package in our improvement plan.

WHAT TO IMPROVE, AND OUR RESPONSE

1 Less wordy, more visual

Give each agent a mascot and a living persona, with some gamification.

Our response: Mascot art for our seven agents, which already have names and jobs, with moods that follow real work: working, waiting for you, done.

2 Research the problem beyond Malaysia

At least at Asia level, and globally if possible, to show that universities everywhere face it.

Our response: A first pass with Malaysian, Asian and global sources is done. Next come interviews with lecturers in a few Asian countries.

3 Make the app and the pitch tally

The flow of the app and the flow of the presentation should match, so the story runs smoothly.

Our response: One storyline, the six steps on page 1, used in the same order by the app, the demo and the slides.

4 Benchmark the mock cohort

The agent that simulates a cohort needs a benchmark. Use the SSRT and CLO surveys we collect every semester.

Our response: Lock the agent's predictions before a semester starts, then score them against that semester's CLO survey and results.

Outcome. Overall, the idea is clear and fits the theme. The travel fellowship is on: Cai Yiyu applied for it on our behalf.